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Code No. 3503

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE) V-Semester (MAIN) Examinations JAN/FEB-2025

NUMBER THEORY AND CRYPTOGRAPHY

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	Write the Fermat's theorem and give example of its use.	2	2	2
b)	What is ECB Block cipher Mode.	2	2	2
c)	Define Kerberos.	2	1	1
d)	What are the features of Secure Electronic Trasaction) (SET).	2	2	2
e)	What is Proxy Firewall.	2	2	3
f)	For SSH packets, what is the advantage, if any, of not including the MAC in the scope of the packet encryption.	2	4	2
2. a)	Define Euclidean algorithm. Write the pseudo code of the algorithm. Use this algorithm to find the greatest common divisor of 2740 and 1760.	6	2	2
b)	Discuss the role of non-repudiation in information security and describe the mechanisms that can be used to achieve non-repudiation in digital communications. Why is non-repudiation critical in legal and financial transactions.	6	3	1
3. a)	Explain the working of DES. How does it overcome the problem of Meet in-the-Middle attack.	6	3	2
b)	Describe the encryption operation using CFB mode of operation. If a bit error occurs in cipher text during transmission, how far does the error propagate in CFB mode. What type of applications use CFB.	6	3	2
4. a)	Explain the concept of digital signatures in public-key cryptography. How do they work, and what are their primary uses in securing digital communication and ensuring data integrity.	6	3	1

(P.T.O.)

b)	How does MD5 work, and what are its primary steps in generating a hash. Give an example.	6	4	2
5. a)	Draw the Pretty Good Privacy (PGP) Message Generation and Message Reception Architecture. Explain.	6	4	2
b)	Explain S/MIME certificate processing method.	6	3	2
6. a)	Describe Transport tunnel modes used for IPsec AH authentication bringing out their scope relevant to IPV4.	6	3	3
b)	What is a firewall. List the characteristics of a good firewall implementation.	6	2	3
7. a)	What are the business requirements for SET. What actions are performed after the merchant receives the purchase request message. Explain with a diagram.	6	2	3
b)	Illustrate with an example the process involved in RSA algorithm.	6	2	1

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